Equity Dilution Burden and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Equity Dilution Burden (EDB), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023a). A value-weighted long/short trading strategy based on EDB achieves an annualized gross (net) Sharpe ratio of 0.55 (0.49), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 21 (20) bps/month with a t-statistic of 2.65 (2.64), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth) is 15 bps/month with a t-statistic of 2.10.

1 Introduction

Production-based asset pricing models predict that firms' investment decisions and production technologies determine the cross-section of expected returns through their impact on firms' marginal value of wealth across states of nature. Despite substantial theoretical development linking production decisions to risk premia, empirical evidence on how specific corporate financing frictions affect expected returns through production channels remains incomplete. We examine whether equity dilution burden (EDB), which measures the extent to which firms rely on equity issuance to finance operations and investment, captures systematic variation in firms' exposure to productivity shocks and investment frictions that manifest in the cross-section of stock returns.

In production-based asset pricing models, firms' optimal investment and financing policies determine their risk exposures through the first-order conditions that equate marginal costs of investment with marginal benefits Cochrane (1991); Zhang (2005). When firms face costly external financing, equity issuance represents a particularly expensive form of capital that firms employ only when internal resources and debt capacity are exhausted, reflecting binding financial constraints that amplify exposure to aggregate productivity shocks Gomes and Schmid (2010). The equity dilution burden signal directly measures this reliance on costly equity financing, capturing cross-sectional variation in firms' marginal costs of investment and their conditional exposure to systematic risk factors.

Firms with high equity dilution burden face elevated adjustment costs that make their investment policies less flexible in response to productivity shocks, creating operating leverage that increases systematic risk Carlson et al. (2004). These adjustment frictions interact with time-varying risk premia in production economies, where firms experiencing negative productivity shocks simultaneously face high marginal utility states when external financing is most costly Livdan et al. (2009). The production-based framework predicts that high-EDB firms, constrained by costly equity financing and unable to adjust investment optimally, will exhibit greater sensitivity to aggregate productivity fluctuations and command higher expected returns.

Furthermore, the timing and persistence of equity issuance decisions create predictable patterns in risk exposures that align with production-based theories of the cross-section Li et al. (2009). Firms issuing equity typically do so when investment opportunities exceed internal resources, but the dilution costs create a wedge between the marginal product of capital and the cost of capital that persists over multiple periods. This persistence in financing frictions translates into persistent differences in conditional risk premia across firms, as those with higher equity dilution burden remain exposed to systematic shocks to investment opportunities and cannot smooth production efficiently Berk et al. (1999).

Our empirical analysis reveals that a value-weighted long/short trading strategy based on EDB achieves an annualized gross Sharpe ratio of 0.55, with the strategy earning average monthly returns of 33 basis points (t-statistic = 4.31). The strategy's performance remains robust after controlling for standard risk factors, generating a monthly alpha of 21 basis points (t-statistic = 2.65) relative to the Fama-French five-factor model augmented with momentum. These results demonstrate economically significant return predictability that persists across different factor model specifications, with alphas ranging from 21 to 37 basis points per month.

The EDB strategy's risk exposures align with production-based theories, exhibiting a significant positive loading of 0.29 (t-statistic = 5.47) on the CMA (conservative minus aggressive) investment factor, consistent with the interpretation that high-EDB firms face binding investment frictions. Implementation considerations strengthen the economic significance of our findings, as the strategy maintains a net Sharpe ratio of 0.49 after accounting for transaction costs, with net alphas remaining statistically significant across all factor models tested. The strategy's performance concentrates in larger, more liquid stocks where implementation is feasible, with the large-cap quintile generating average returns of 22 basis points per month (t-statistic = 2.52).

Controlling for closely related anomalies reveals that EDB contains distinct information about cross-sectional return variation linked to production frictions. When we control simultaneously for six closely related signals including share issuance, net payout yield, and asset growth, plus the Fama-French six factors, the EDB strategy maintains an alpha of 15 basis points per month (t-statistic = 2.10). This residual predictability suggests that EDB captures unique aspects of how equity financing frictions affect firms' production decisions and risk exposures beyond what existing measures of investment and financing activities explain.

Our study contributes to the production-based asset pricing literature by providing direct evidence that equity dilution burden, as a measure of costly external financing, predicts cross-sectional returns through channels consistent with investment frictions and conditional risk premia. While Pontiff and Woodgate (2008) document that share issuance predicts returns and Daniel and Titman (2006) show that financing activities relate to return predictability, our EDB measure more precisely captures the ongoing burden of equity financing on firms' production decisions and risk exposures. Unlike Cooper et al. (2008) who focus on asset growth as a comprehensive measure of investment, EDB specifically isolates the financing friction component that production-based models identify as crucial for understanding risk premia.

Methodologically, we advance the literature by demonstrating that EDB's predictive power survives stringent tests including controls for transaction costs and simultaneous inclusion of multiple related anomalies, addressing concerns raised by Novy-Marx and Velikov (2016a) about the implementability of many documented anomalies. Our comprehensive analysis using the Novy-Marx and Velikov (2023b) protocol ensures that EDB represents a robust addition to the factor zoo rather than a spurious pattern. The finding that EDB maintains significant alpha even after controlling for established factors like CMA and RMW from Fama and French (2015) suggests that existing factor models incompletely capture the risk associated with equity financing frictions.

The broader implications of our findings extend to understanding how corporate financing decisions interact with production technologies to determine equilibrium risk premia. The persistence and magnitude of the EDB premium, particularly among large-cap stocks where implementation is feasible, challenge purely risk-based explanations that rely on rare disasters or consumption-based factors while supporting production-based theories that emphasize investment frictions Zhang (2005). Our results suggest that incorporating measures of financing frictions like EDB into asset pricing models could improve their ability to explain cross-sectional return variation and provide insights into how production decisions and financial constraints jointly determine firms' systematic risk exposures in equilibrium.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of annual observations. We focus on non-financial firms and exclude observations with missing or negative values for the key components of our signal construction.Equity Dilution Burden signal relies on two fundamental accounting variables. Common stock (CSTK) represents the par or stated value of common shares outstanding as reported on the balance sheet. This item reflects the nominal value assigned to issued common equity, typically recorded at a fixed par value per share multiplied by the number of shares issued. Total liabilities (LT) captures the sum of all current and long-term obligations owed by the firm, representing the aggregate claims of creditors on the firm's assets.construct the Equity Dilution Burden signal by computing the year-over-year change in common stock scaled by lagged total liabilities: (CSTK(t) - CSTK(t-1)) / LT(t-1). This measure captures the extent to which firms issue new common equity relative to their existing debt obligations. A positive value indicates an increase in the par value of common stock outstanding, typically resulting from new equity issuances, while the scaling by lagged liabilities normalizes this change relative to the firm's debt burden. We calculate this signal using fiscal year-end values to ensure consistency across firms with different fiscal year conventions. We require non-missing values for both CSTK and LT in consecutive years and exclude observations where lagged liabilities equal zero to avoid undefined ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDB signal. Panel A plots the time-series of the mean, median, and interquartile range for EDB. On average, the cross-sectional mean (median) EDB is -0.03 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EDB data. The signal's interquartile range spans -0.01 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EDB signal for the CRSP universe. On average, the EDB signal is available for 6.37% of CRSP names, which on average make up 7.67% of total market capitalization.

4 Does EDB predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDB using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDB portfolio and sells the low EDB portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short EDB strategy earns an average return of 0.33% per month with a t-statistic of 4.31. The annualized Sharpe ratio of the strategy is 0.55. The alphas range from 0.21% to 0.37% per month and have t-statistics exceeding 2.65 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.29, with a t-statistic of 5.47 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 542 stocks and an average market capitalization of at least $1,474 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the decile sort using NYSE breakpoints and value-weighted portfolios, and equals 23 bps/month with a t-statistics of 2.51. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-one exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016b). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 19-30bps/month. The lowest return, (19 bps/month), is achieved from the decile sort using NYSE breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.05. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EDB trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-one cases.

Table 3 provides direct tests for the role size plays in the EDB strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDB, as well as average returns and alphas for long/short trading EDB strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the EDB strategy achieves an average return of 22 bps/month with a t-statistic of 2.52. Among these large cap stocks, the alphas for the EDB strategy relative to the five most common factor models range from 15 to 23 bps/month with t-statistics between 1.61 and 2.60.

5 How does EDB perform relative to the zoo?

Figure 2 puts the performance of EDB in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the EDB strategy falls in the distribution. The EDB strategy's gross (net) Sharpe ratio of 0.55 (0.49) is greater than 93% (98%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EDB strategy (red line).[2] Ignoring trading costs, a \$1 invested in the EDB strategy would have yielded \$8.72 which ranks the EDB strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EDB strategy would have yielded \$6.37 which ranks the EDB strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

of gross and Novy-Marx and Velikov (2016b) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EDB relative to those. Panel A shows that the EDB strategy gross alphas fall between the 65 and 70 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDB strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDB ranks between the 82 and 87 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EDB add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EDB with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDB or at least to weaken the power EDB has predicting the cross-section of returns. Figure 7 plots histograms

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

of t-statistics for predictability tests of EDB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDB} EDB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDB. Stocks are finally grouped into five EDB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDB and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDB signal in these Fama-MacBeth regressions exceed 1.57, with the minimum t-statistic occurring when controlling for Net Payout Yield. Controlling for all six closely related anomalies, the t-statistic on EDB is 0.70.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDB strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDB strategy earns alphas that range from 14-23bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.81, which is achieved when controlling for Net Payout Yield. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDB trading strategy achieves an alpha of 15bps/month with a t-statistic of 2.10.

7 Does EDB add relative to the whole zoo?

Finally, we can ask how much adding EDB to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EDB signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes EDB grows to $2563.07.

8 Conclusion

This study provides compelling evidence that Equity Dilution Burden (EDB) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that EDB-based trading strategies generate substantial

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EDB is available.

risk-adjusted returns even after accounting for transaction costs and controlling for well-established risk factors.

The key findings reveal that a value-weighted long/short strategy based on EDB delivers an impressive annualized net Sharpe ratio of 0.49, with monthly abnormal returns of 20 basis points relative to the Fama-French five-factor model augmented with momentum. The statistical significance of these results (t-statistic of 2.64) underscores the reliability of EDB as a return predictor. Particularly noteworthy is the signal's ability to generate alpha even when controlling for six closely related equity issuance and growth-based strategies, producing a monthly alpha of 15 basis points with a t-statistic of 2.10. This finding suggests that EDB captures unique information about firm value destruction through equity dilution that is not fully reflected in existing factor models or related signals.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For portfolio managers, EDB offers a implementable strategy that survives realistic transaction costs, as evidenced by the modest difference between gross and net returns. The signal's orthogonality to existing factors suggests it could serve as a valuable addition to multi-factor models and quantitative investment strategies. Furthermore, the economic intuition underlying EDB—that excessive equity dilution imposes costs on existing shareholders—provides a clear theoretical foundation for its predictive power.

Several limitations of this study merit consideration. First, while we control for numerous related factors, the possibility remains that EDB's predictive power may be partially explained by other unidentified risk factors or market frictions. Second, the analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question. Third, the stability of the EDB signal's performance across different market regimes and its behavior during extreme market conditions warrant further investigation.

Future research should explore several promising avenues. First, examining the international evidence on EDB could provide valuable insights into whether the signal's effectiveness is a global phenomenon or specific to certain market structures. Second, investigating the underlying economic mechanisms through which equity dilution affects future returns—whether through agency costs, market timing, or information asymmetries—would enhance our theoretical understanding. Third, analyzing the interaction between EDB and corporate governance characteristics could reveal when dilution is most value-destructive. Finally, exploring the signal's performance in different asset classes, such as convertible bonds or REITs where dilution mechanics differ, could broaden the applicability of these findings. As markets evolve and new forms of equity-linked securities emerge, understanding the pricing implications of dilution will remain a critical area of financial research.

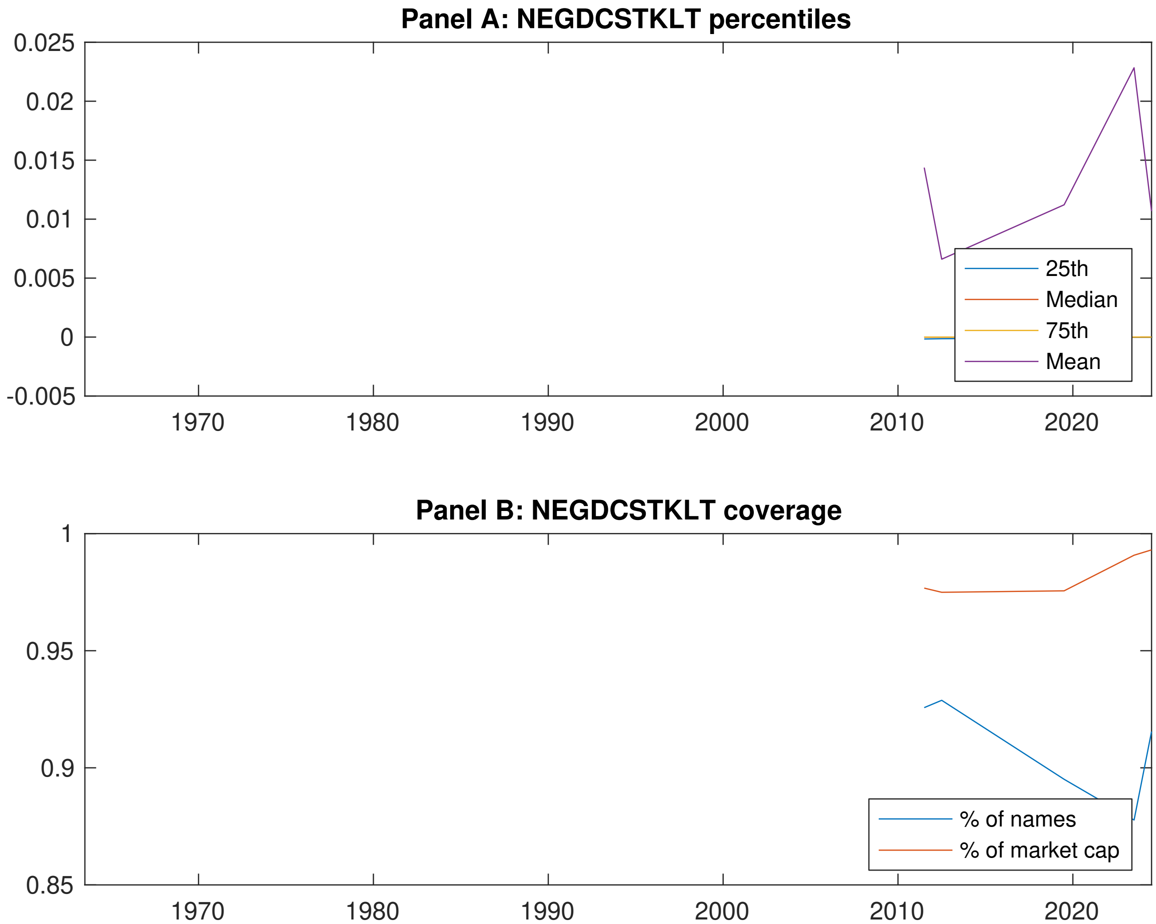


Figure 1: Times series of EDB percentiles and coverage.
This figure plots descriptive statistics for EDB. Panel A shows cross-sectional percentiles of EDB over the sample. Panel B plots the monthly coverage of EDB relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDB. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EDB-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.41	0.55	0.60	0.69	0.74	0.33
	[2.42]	[3.17]	[3.39]	[4.24]	[4.68]	[4.31]
α_{CAPM}	-0.16	-0.05	-0.01	0.13	0.20	0.37
	[-3.23]	[-1.33]	[-0.30]	[2.85]	[4.36]	[4.75]
α_{FF3}	-0.15	-0.04	-0.02	0.09	0.16	0.31
	[-2.91]	[-1.00]	[-0.50]	[2.00]	[3.67]	[4.10]
α_{FF4}	-0.12	-0.04	-0.01	0.07	0.15	0.27
	[-2.22]	[-0.95]	[-0.14]	[1.52]	[3.36]	[3.47]
α_{FF5}	-0.16	0.00	-0.03	0.01	0.08	0.23
	[-3.00]	[0.00]	[-0.59]	[0.22]	[1.80]	[3.07]
α_{FF6}	-0.13	-0.00	-0.01	-0.00	0.08	0.21
	[-2.44]	[-0.06]	[-0.29]	[-0.01]	[1.75]	[2.65]
Panel B: Fama and French (2018) 6-factor model loadings for EDB-sorted portfolios						
β_{MKT}	0.97	1.01	1.05	1.01	0.97	0.01
	[77.03]	[104.34]	[98.66]	[97.99]	[91.50]	[0.43]
β_{SMB}	-0.01	0.03	0.00	-0.07	-0.02	-0.01
	[-0.43]	[2.16]	[0.14]	[-4.73]	[-1.15]	[-0.37]
β_{HML}	-0.01	-0.01	0.02	0.08	0.03	0.04
	[-0.50]	[-0.73]	[1.10]	[4.04]	[1.61]	[1.26]
β_{RMW}	0.08	-0.07	0.03	0.11	0.14	0.05
	[3.41]	[-3.90]	[1.31]	[5.47]	[6.61]	[1.49]
β_{CMA}	-0.09	-0.07	-0.01	0.19	0.19	0.29
	[-2.64]	[-2.66]	[-0.41]	[6.44]	[6.42]	[5.47]
β_{UMD}	-0.04	0.00	-0.02	0.01	0.00	0.04
	[-3.23]	[0.37]	[-1.79]	[1.32]	[0.15]	[2.27]
Panel C: Average number of firms (n) and market capitalization (me)						
n	812	670	542	667	734	
me ($\$10^6$)	1709	1474	2100	2300	2534	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EDB strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016b) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.33	0.37	0.31	0.27	0.23	0.21
			[4.31]	[4.75]	[4.10]	[3.47]	[3.07]	[2.65]
Quintile	NYSE	EW	0.49	0.58	0.48	0.40	0.33	0.27
			[7.00]	[8.64]	[8.06]	[6.78]	[5.89]	[4.92]
Quintile	Name	VW	0.30	0.32	0.27	0.24	0.22	0.21
			[3.81]	[4.11]	[3.46]	[3.10]	[2.81]	[2.60]
Quintile	Cap	VW	0.25	0.28	0.24	0.20	0.20	0.17
			[3.44]	[3.76]	[3.22]	[2.69]	[2.69]	[2.33]
Decile	NYSE	VW	0.23	0.25	0.17	0.14	0.15	0.13
			[2.51]	[2.70]	[1.90]	[1.50]	[1.70]	[1.43]
Panel B: Net Returns and Novy-Marx and Velikov (2016b) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.30	0.33	0.27	0.25	0.21	0.20
			[3.82]	[4.25]	[3.62]	[3.28]	[2.86]	[2.64]
Quintile	NYSE	EW	0.29	0.39	0.30	0.26	0.15	0.13
			[3.88]	[5.40]	[4.59]	[4.00]	[2.51]	[2.16]
Quintile	Name	VW	0.26	0.28	0.23	0.21	0.19	0.19
			[3.32]	[3.60]	[2.98]	[2.80]	[2.52]	[2.42]
Quintile	Cap	VW	0.22	0.24	0.20	0.18	0.17	0.16
			[2.94]	[3.27]	[2.73]	[2.45]	[2.43]	[2.24]
Decile	NYSE	VW	0.19	0.20	0.13	0.11	0.12	0.10
			[2.05]	[2.16]	[1.42]	[1.22]	[1.30]	[1.17]

Table 3: Conditional sort on size and EDB

This table presents results for conditional double sorts on size and EDB. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDB. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDB and short stocks with low EDB .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		EDB Quintiles					EDB Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.43	0.67	0.90	0.92	0.98	0.58	0.66	0.58	0.51	0.40	0.36
		[1.63]	[2.61]	[3.69]	[3.71]	[4.36]	[6.18]	[7.24]	[6.85]	[6.03]	[5.01]	[4.45]
	(2)	0.52	0.76	0.86	0.94	0.92	0.40	0.49	0.36	0.32	0.27	0.24
		[2.20]	[3.21]	[3.75]	[4.27]	[4.35]	[4.38]	[5.45]	[4.51]	[3.94]	[3.32]	[2.95]
	(3)	0.57	0.62	0.74	0.90	0.89	0.33	0.38	0.28	0.25	0.21	0.19
		[2.72]	[2.92]	[3.45]	[4.42]	[4.63]	[4.01]	[4.75]	[3.82]	[3.26]	[2.84]	[2.45]
	(4)	0.50	0.57	0.83	0.78	0.79	0.29	0.34	0.24	0.23	0.06	0.07
		[2.59]	[2.90]	[4.16]	[4.19]	[4.39]	[3.56]	[4.18]	[3.22]	[3.08]	[0.89]	[1.04]
	(5)	0.46	0.49	0.47	0.60	0.68	0.22	0.23	0.19	0.16	0.18	0.15
		[2.80]	[2.73]	[2.72]	[3.58]	[4.35]	[2.52]	[2.60]	[2.13]	[1.69]	[1.93]	[1.61]

Panel B: Portfolio average number of firms and market capitalization

		EDB Quintiles Average n					EDB Quintiles Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	383	382	381	381	374	31	34	40	30	29
	(2)	109	107	108	108	106	57	56	58	56	55
	(3)	78	78	77	77	78	97	96	98	100	99
	(4)	65	65	65	65	65	201	205	214	214	216
	(5)	60	60	60	60	60	1449	1494	1640	1660	1891

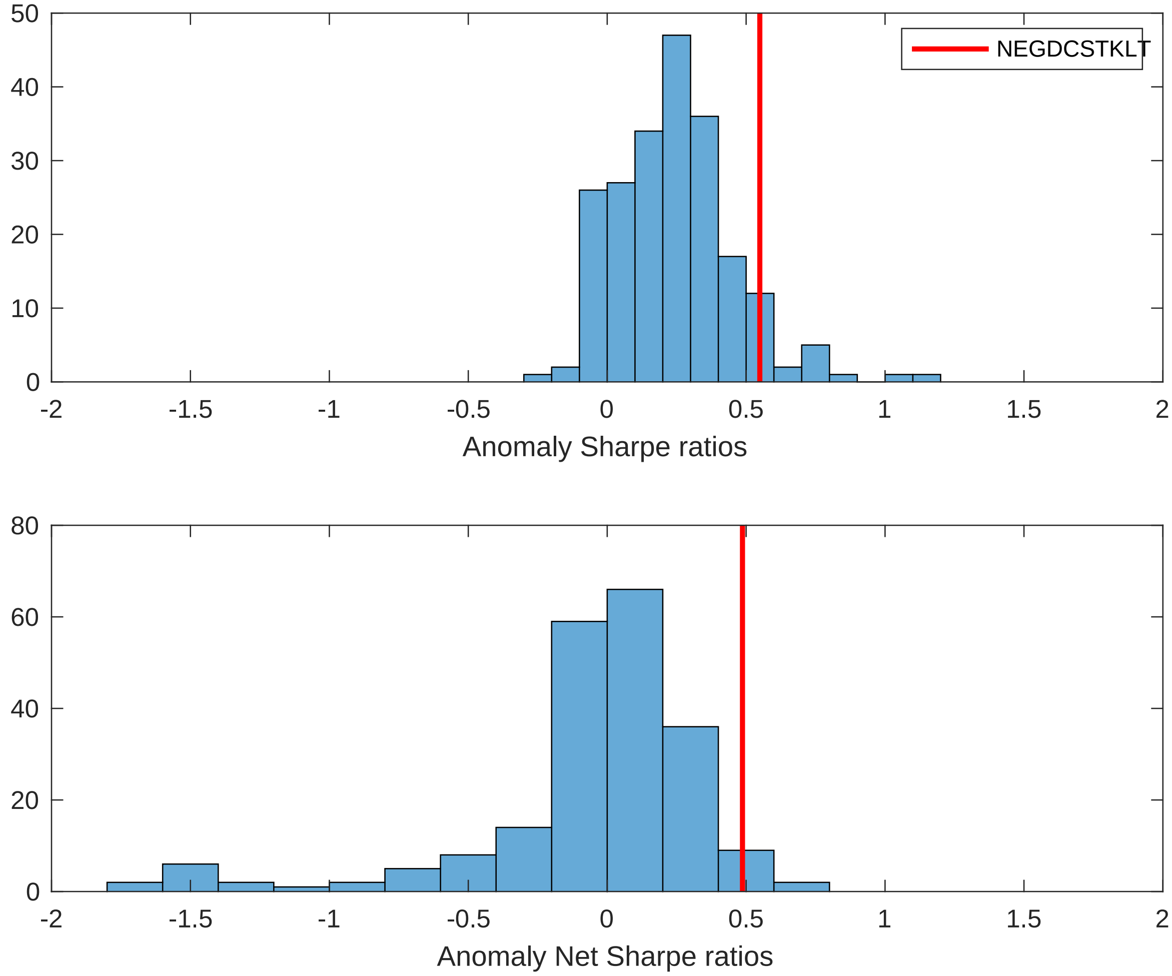


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDB with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

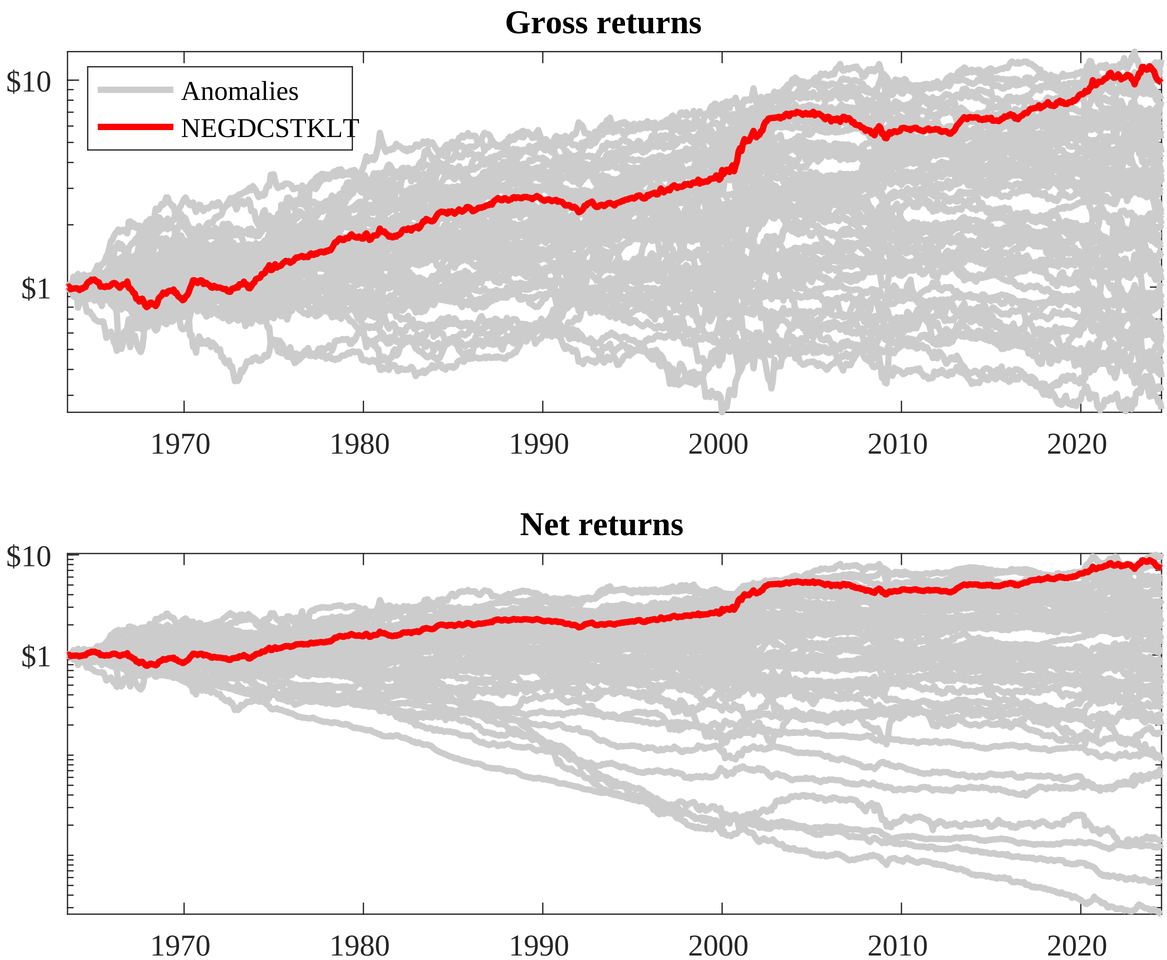
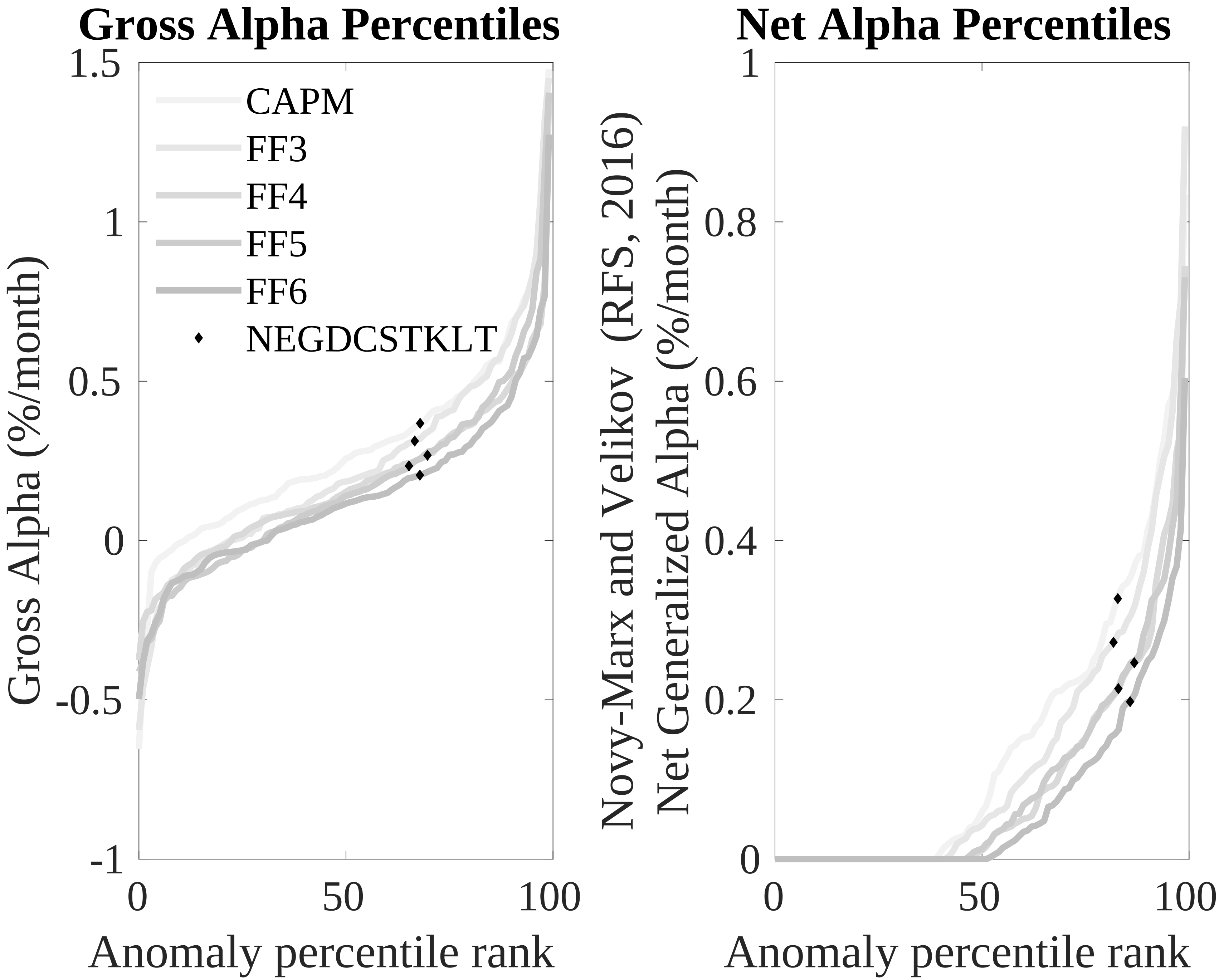


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDB trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDCSTKLT
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

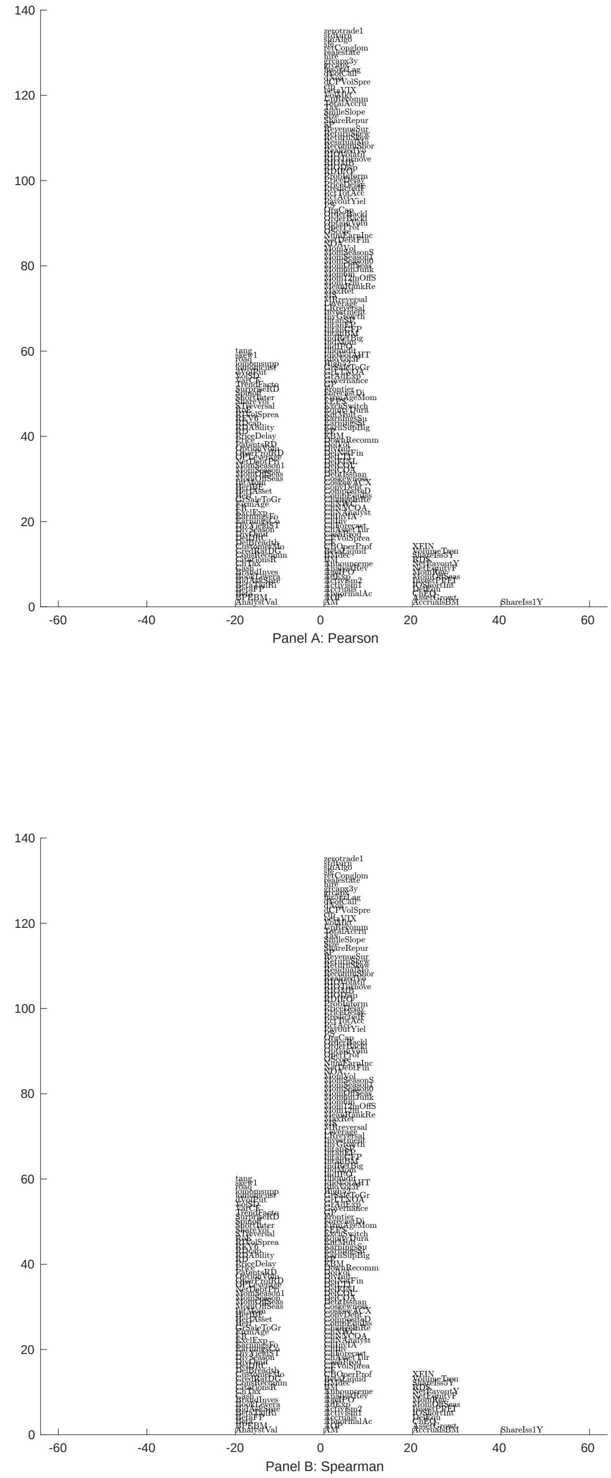


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with EDB. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

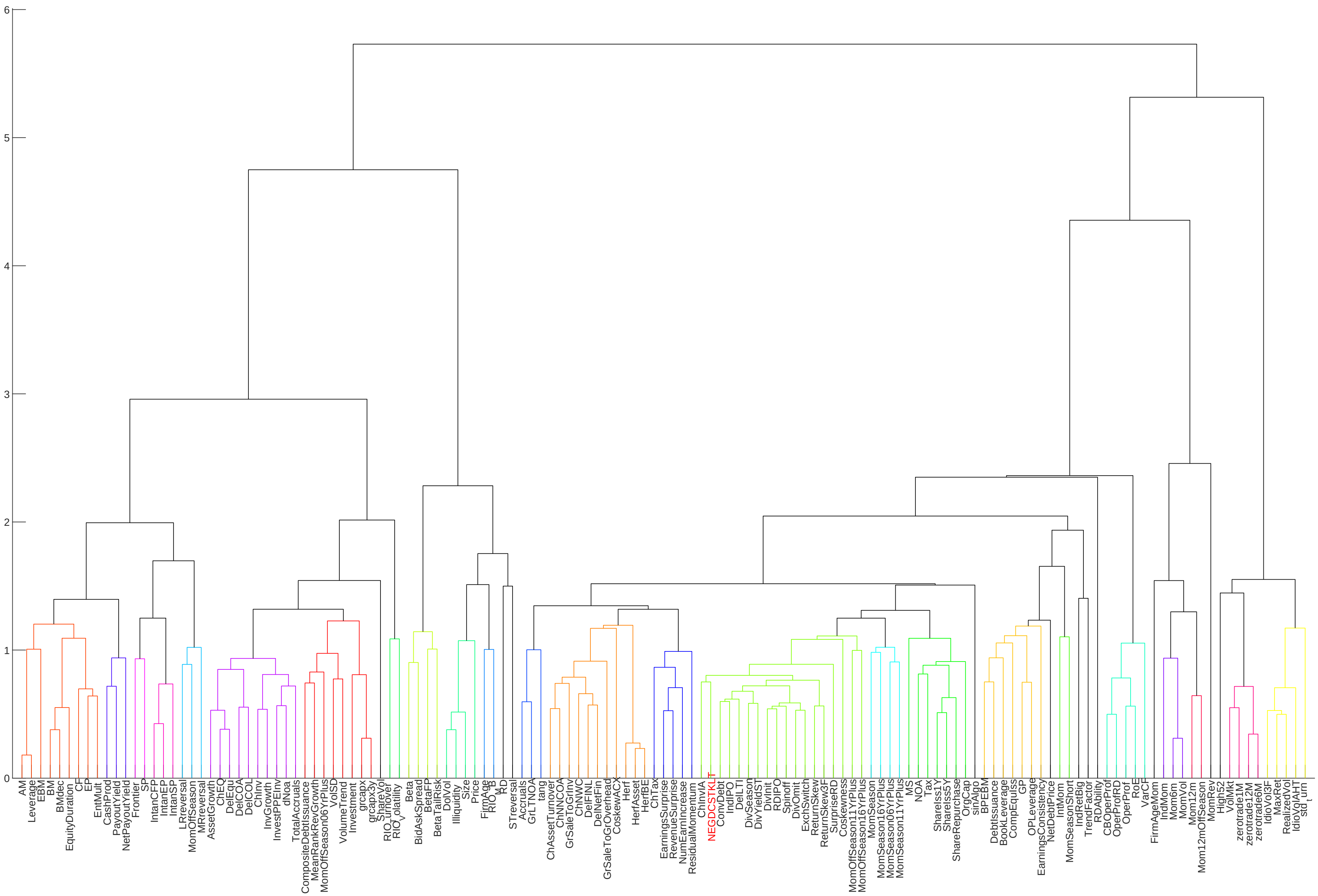


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

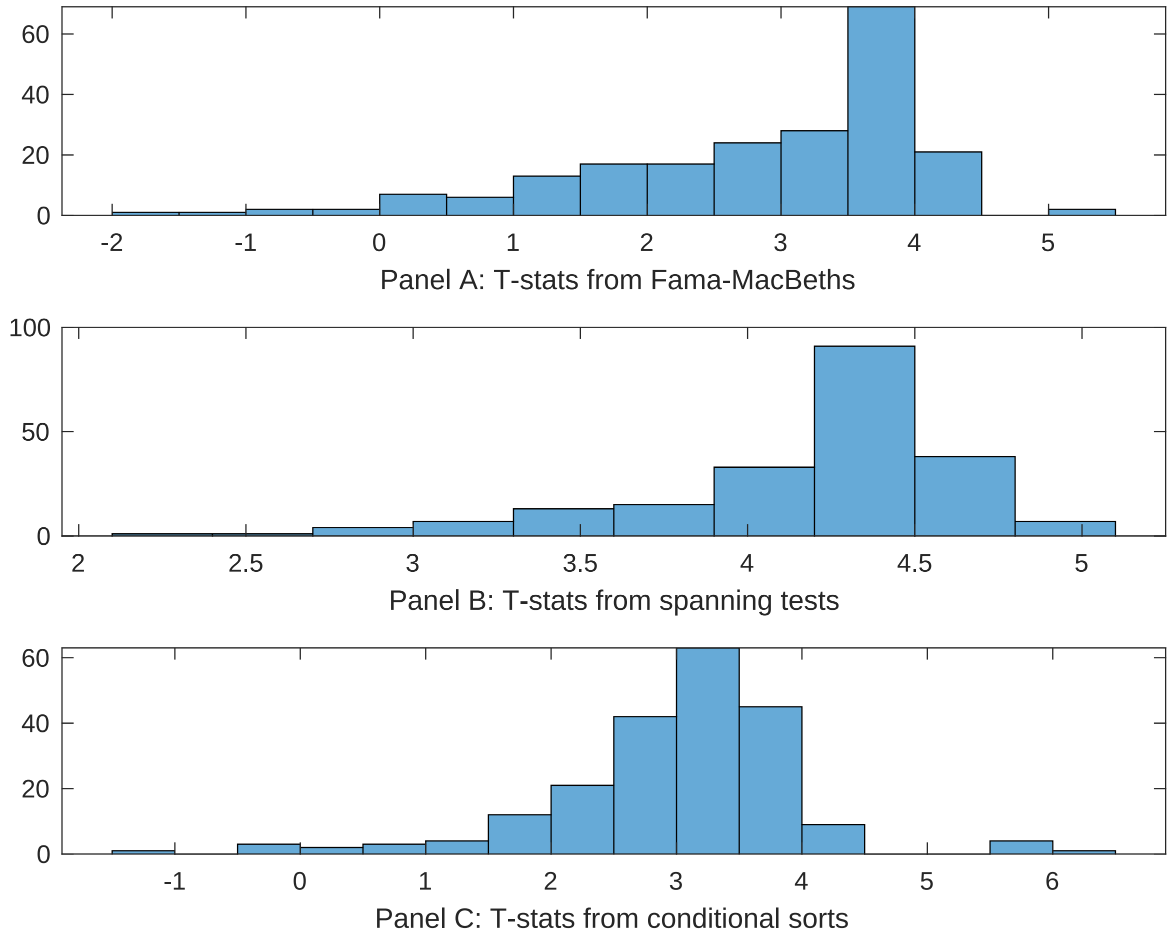


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of EDB conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDB} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDB} EDB_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDB,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDB. Stocks are finally grouped into five EDB portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDB trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on EDB. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDB} EDB_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13	0.12	0.17	0.13	0.12	0.13	0.12
	[6.06]	[5.42]	[6.92]	[6.45]	[5.70]	[6.16]	[3.95]
EDB	0.23	0.14	0.21	0.20	0.26	0.21	0.54
	[2.96]	[1.57]	[2.44]	[2.78]	[2.89]	[2.34]	[0.70]
Anomaly 1	0.14						0.69
	[6.13]						[2.50]
Anomaly 2		0.21					0.92
		[1.90]					[0.87]
Anomaly 3			0.40				-0.12
			[3.40]				[-0.53]
Anomaly 4				0.27			0.24
				[4.88]			[4.23]
Anomaly 5					0.11		-0.79
					[3.19]		[-0.13]
Anomaly 6						0.94	0.64
						[8.02]	[5.40]
# months	715	715	720	715	720	720	715
$\bar{R}^2$(%)	0	1	0	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the EDB trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDB} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Share issuance (1 year), Net Payout Yield, Growth in book equity, Share issuance (5 year), Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.18	0.23	0.18	0.14	0.20	0.20	0.15
	[2.48]	[3.08]	[2.52]	[1.81]	[2.67]	[2.57]	[2.10]
Anomaly 1	27.79						12.84
	[7.38]						[2.85]
Anomaly 2		18.26					6.19
		[6.28]					[1.97]
Anomaly 3			34.18				36.74
			[8.60]				[6.40]
Anomaly 4				25.87			11.04
				[6.45]			[2.47]
Anomaly 5					17.31		-5.09
					[4.60]		[-0.96]
Anomaly 6						-1.57	-20.58
						[-0.34]	[-4.17]
mkt	1.63	3.77	2.06	2.12	0.63	0.95	3.97
	[0.92]	[2.05]	[1.18]	[1.18]	[0.35]	[0.52]	[2.25]
smb	1.29	3.18	-0.93	1.91	-0.01	0.34	3.03
	[0.50]	[1.20]	[-0.37]	[0.73]	[-0.00]	[0.13]	[1.20]
hml	4.99	-1.41	1.53	7.83	3.45	5.85	0.56
	[1.46]	[-0.38]	[0.45]	[2.26]	[0.98]	[1.65]	[0.16]
rmw	-9.57	-5.17	7.10	-3.79	7.06	5.50	-7.53
	[-2.38]	[-1.33]	[2.08]	[-1.00]	[1.99]	[1.54]	[-1.80]
cma	10.73	12.63	-5.14	15.41	10.54	29.88	3.96
	[1.94]	[2.25]	[-0.82]	[2.85]	[1.65]	[3.93]	[0.53]
umd	3.04	5.76	3.70	2.62	4.73	4.30	2.07
	[1.72]	[3.22]	[2.13]	[1.46]	[2.63]	[2.35]	[1.20]
# months	728	728	732	728	732	732	728
$\bar{R}^2$(%)	17	16	19	16	14	11	25

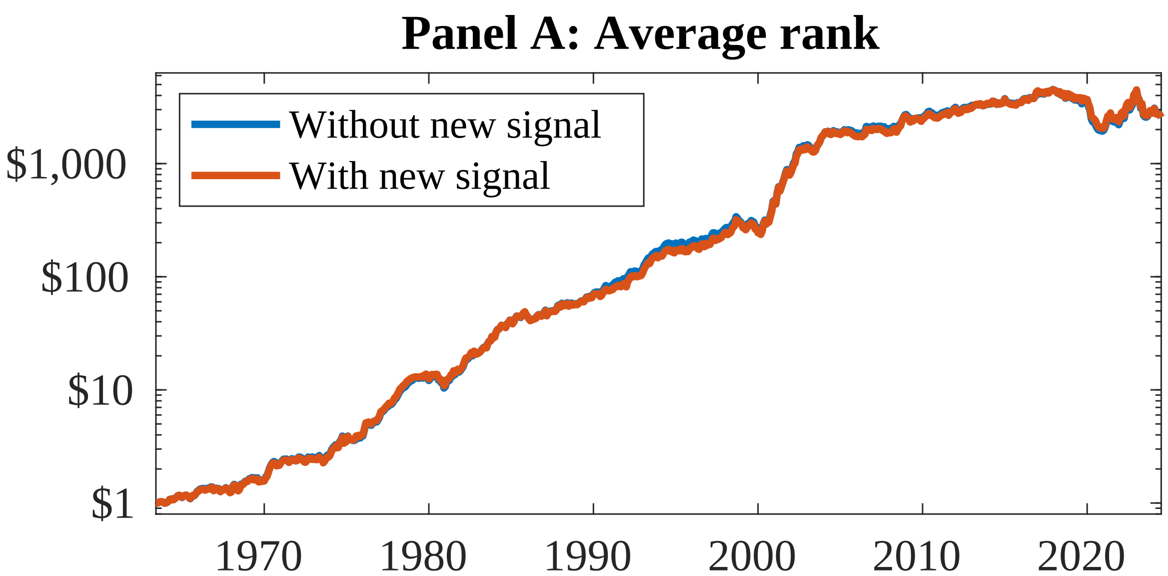


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EDB. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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